

 Directional Drilling & MLWD Services

Halliburton Drilling Systems Indonesia

June 2026

Safety Moment

HSE & SQ MOMENT

Line of Fire

Be careful when positioning your body!
Stay out of the **line of fire.**

- 1** Keep your hands away from hot or extremely cold surfaces.
- 2** Watch out for unexpected object movements. Keep all parts of your body out of the path of objects.
- 3** Don't position yourself between vehicles or moving objects.
- 4** Respect the DROP Zone and use accessories to prevent dropped objects.
- 5** Apply LOTO to valves, electrical panels, motors, etc., whenever necessary.



Safe Zone



If you have any questions about Health, Safety, Environment or about any Process, use your Stop Work Authority (SWA), and contact your Supervisor or the HSE and Service Quality Department.

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Sperry Drilling Value Proposition

We **COLLABORATE** to **DELIVER**
ENGINEERED DRILLING SOLUTIONS
and **RESERVOIR INSIGHT**
to **MAXIMIZE ASSET VALUE**

MAXIMIZE ASSET VALUE

Drill to produce | Enhance reservoir understanding | Reduce well time

COLLABORATE

Listen for value drivers | Solution-based designs | Integrate for value

Why do we need directional drilling?

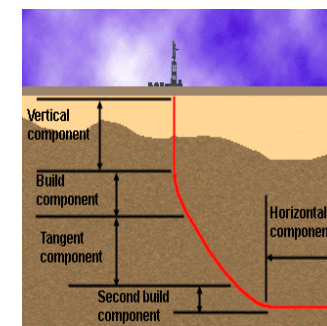
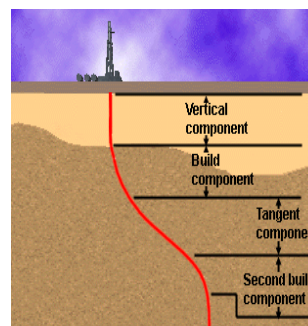
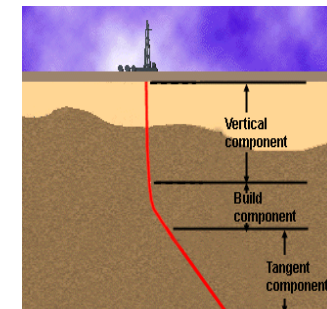
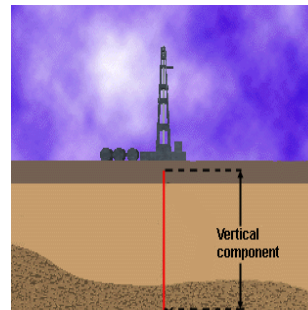


Directional drilling provides a tangible and significant benefit to the client including:

- Reduced costs per barrel
- Improved recovery from the well

Applications for directional drilling include:

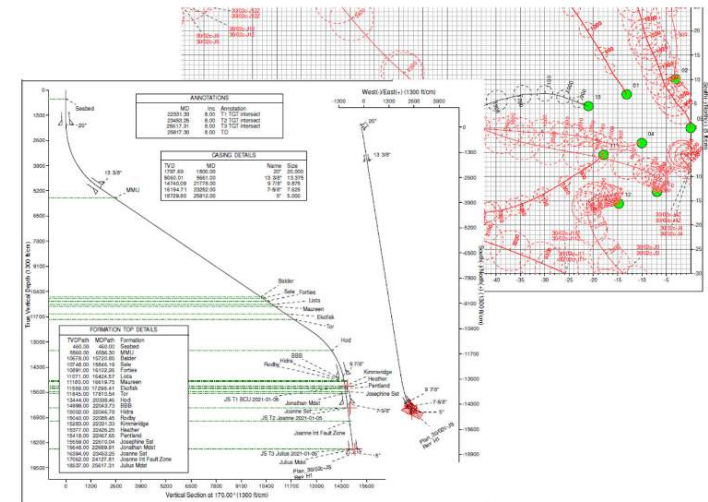
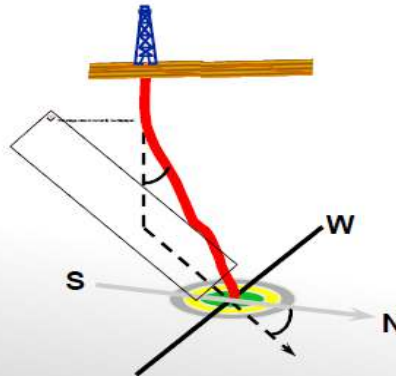
- Inaccessible locations
- Sidetrack a vertical well bore
- Drill different type of reservoir
- Fault control
- Relief well



Deviated Well

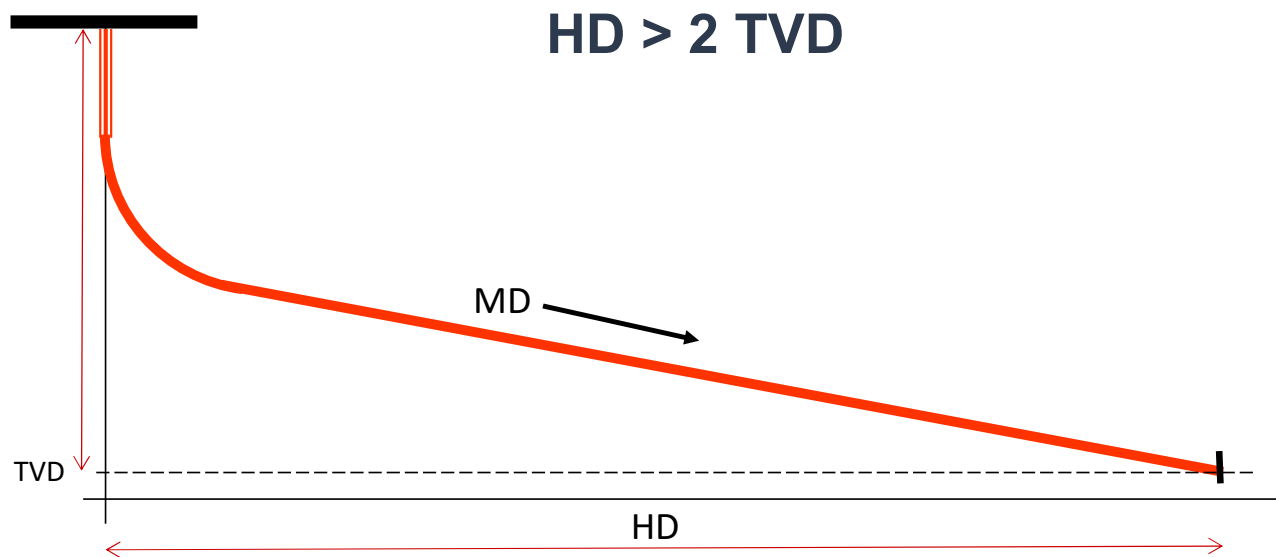
► What is Basic Directional Drilling?

- Procedure and method of deviating a well bore along a planned course to a subsurface target
 - Location is a given lateral distance and direction from the vertical
- **Deviated wells are drilled with intentional control to hit a pre-determined target using:**
 - Inclination (angle)
 - Azimuth (direction)
 - Vertical well → Inc < 3°
 - Horizontal well → Inc > 86°

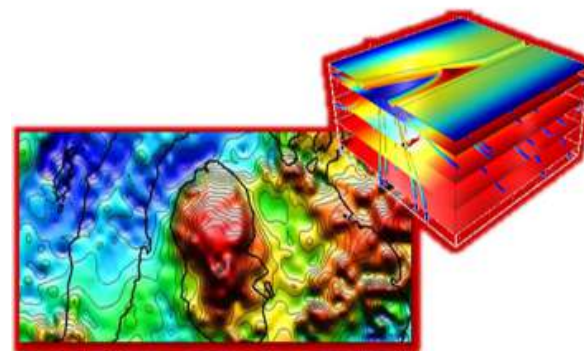
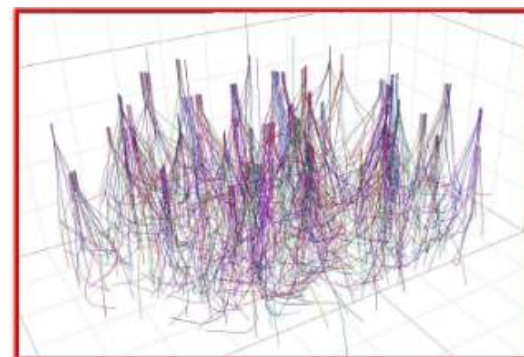
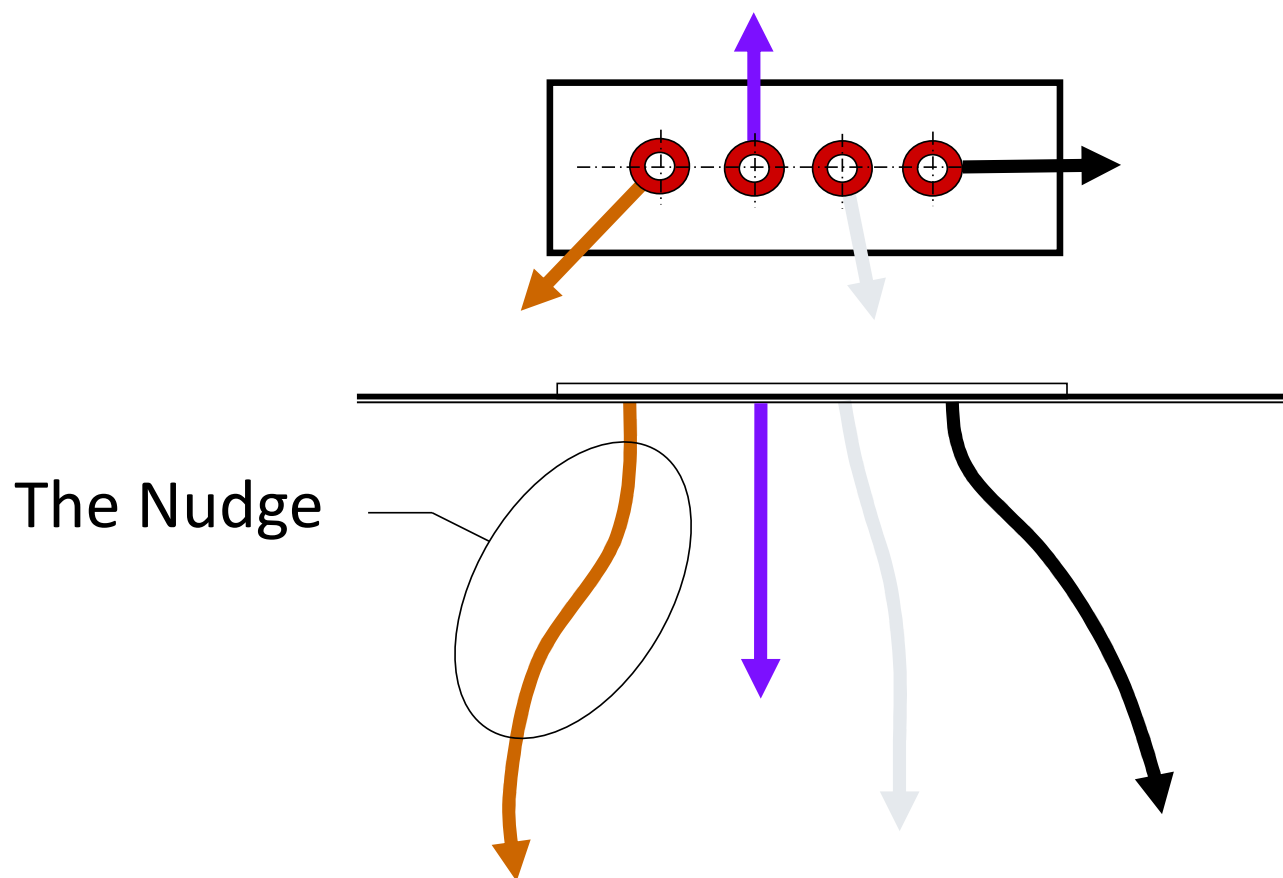


Extended Reach Drilling (ERD)

- The drilling of a wellbore in which the final horizontal displacement greater than twice the vertical depth



Nudging of Wells for Better Shallow Depth Separation



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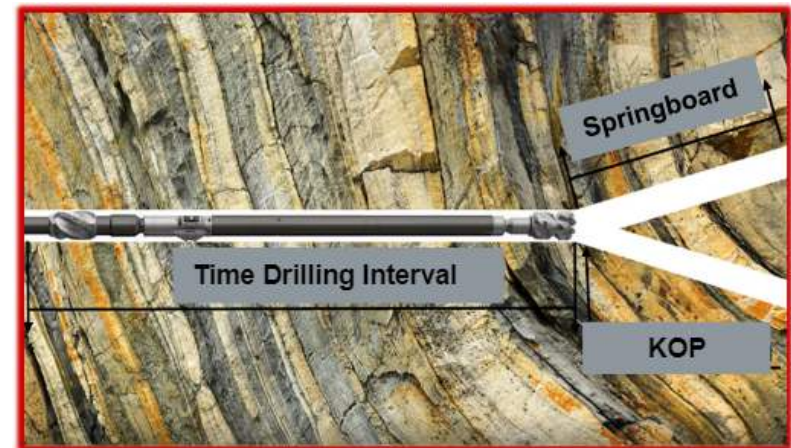
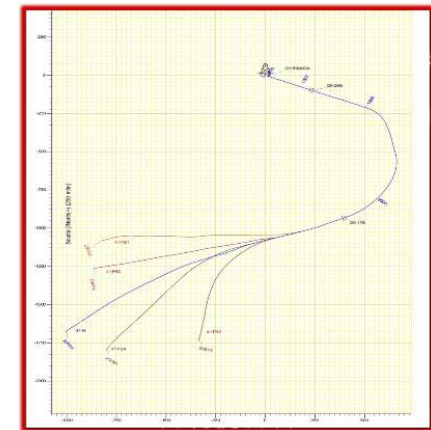
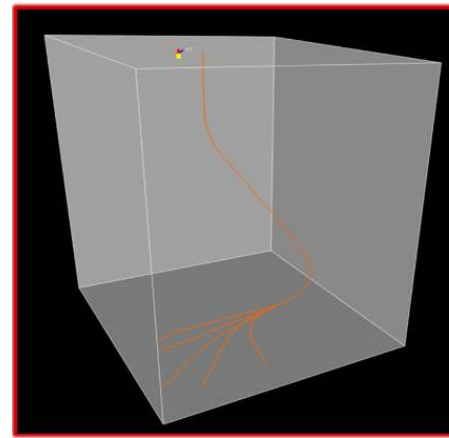
Planning Directional Wells



- New wells
- Existing wells
 - Deepening
 - Sidetracking
 - Going horizontal

Well Profile

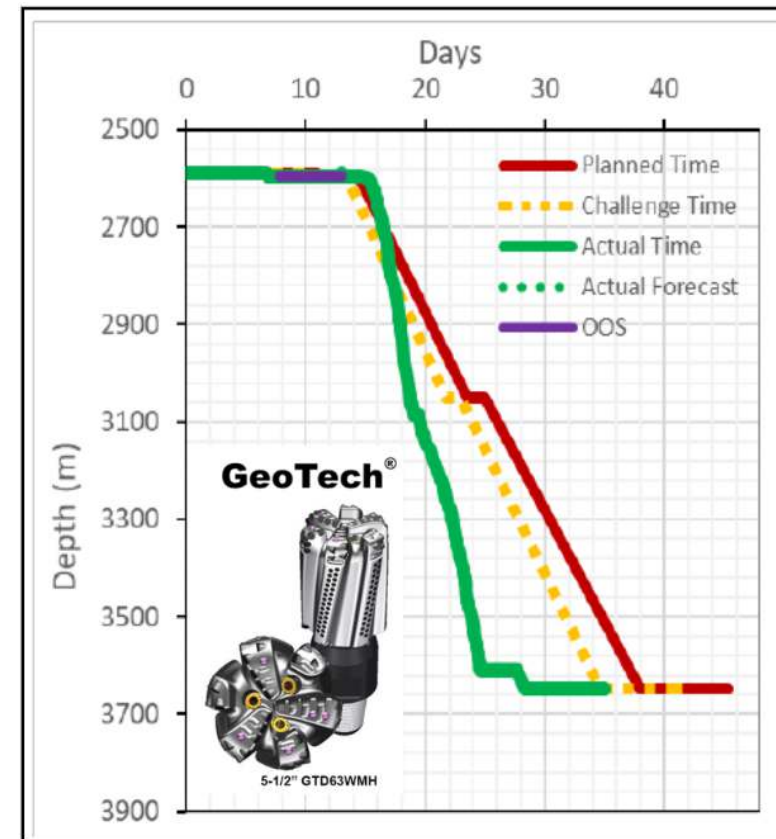
- The trajectory - or well path - from the surface location through the target(s)
- Basic design data :
 - kick-off depth
 - vertical depth of target
 - horizontal displacement to target from surface location



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Considerations

- Avoiding nearby wells is a major concern
- Depth of kick-off affects the profile
- Shape of trajectory : keep it simple
- Curvature, build-, drop-, turn rates should be at minimum
- Length and inclination of tangent section(s)
- Target intersection details
- Length and inclination of horizontal section



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Selection Of Kick-Off Point



- Considerations for a planned kick-off point
 - Formation properties (possible washouts)
 - Inclination and direction (use of gyro or single shot, MWD)
 - Directional assembly performance (RSS, Motor)
- If it's a cased hole sidetrack, surveys, CBL & CCL are required
 - Section milling
 - Whipstock + window milling
- Open hole sidetrack / kick-off
 - Distance from casing shoe is important (Magnetic interference)
 - A cement plug may be required
 - An open hole Whipstock is optional

Minimum Build Rate



- Minimum BUR is defined by a circular arc from the KOP to the target entry point
- Minimum 15° inclination in the tangent section for a stable direction

Maximum Build Rate



Concerns :

- Casing / liner running must be possible
- Casing /liner bending stress must remain within pipe strength limits
- Mechanical loads imposed on drillstring elements (torque, drag, fatigue) or casing (wear, tensile strength reduction due to bending stress in doglegs)
- Maximum curvature for surveying and logging tools
- Completion requirements
- Risk of casing wear or keyseat in open hole

Why Large Direction Changes Are Undesirable

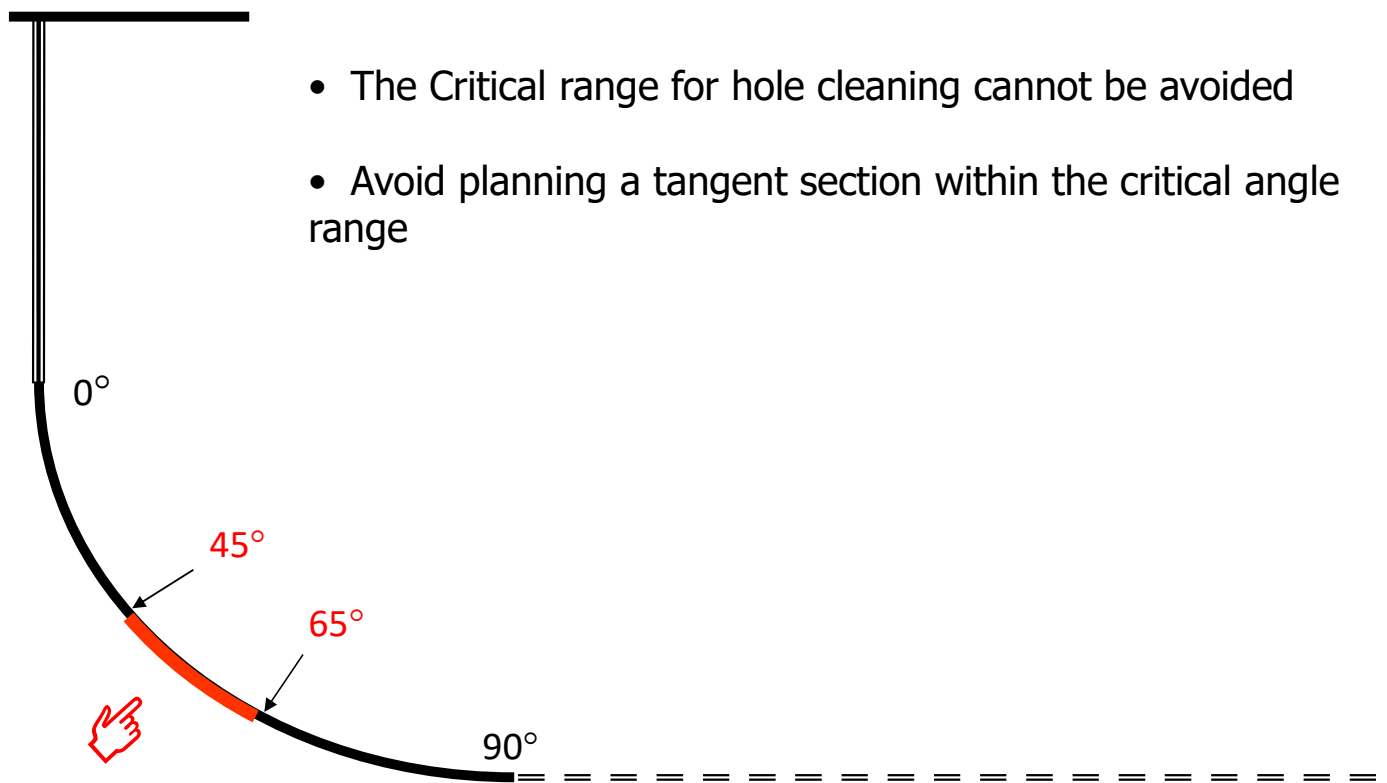


- With increasing inclination, it drastically increases the DLS
- Increases the torque and drag
- Increases the time required for directional work
- Reduces the angle building capability of a given directional tool

Critical Hole Angle For Cleaning

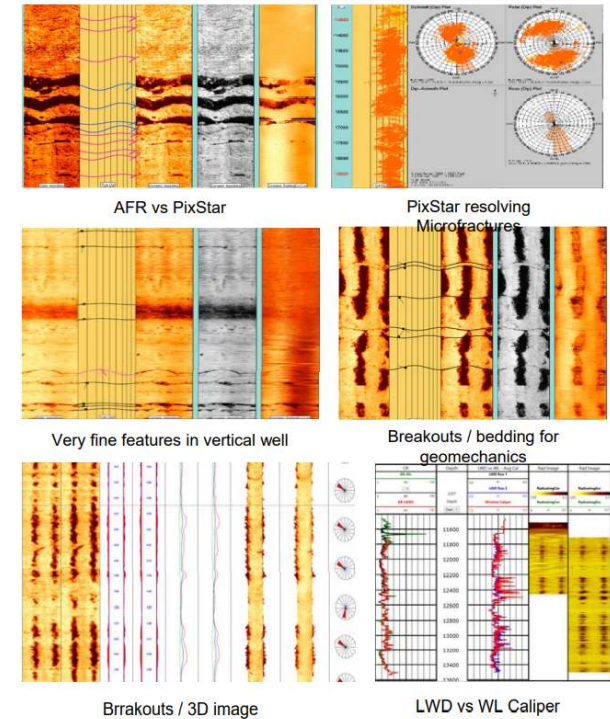


- The Critical range for hole cleaning cannot be avoided
- Avoid planning a tangent section within the critical angle range



Limitations

- Formation drillability
- Borehole stability
- Steering ability, flexibility of drillstring
- Drillpipe strength
- Hole cleaning, cutting transportation
- Standpipe / Pump pressure, ECD
- Rig capacity (hook load, rotary torque, pumps, shakers)



Directional Well Types



Long radius	2-6°/100 ft
Medium radius	6-35°/100 ft
Intermediate radius	35-65°/100 ft
Short radius	65-115°/100 ft
Ultra short radius	> 115°/100ft

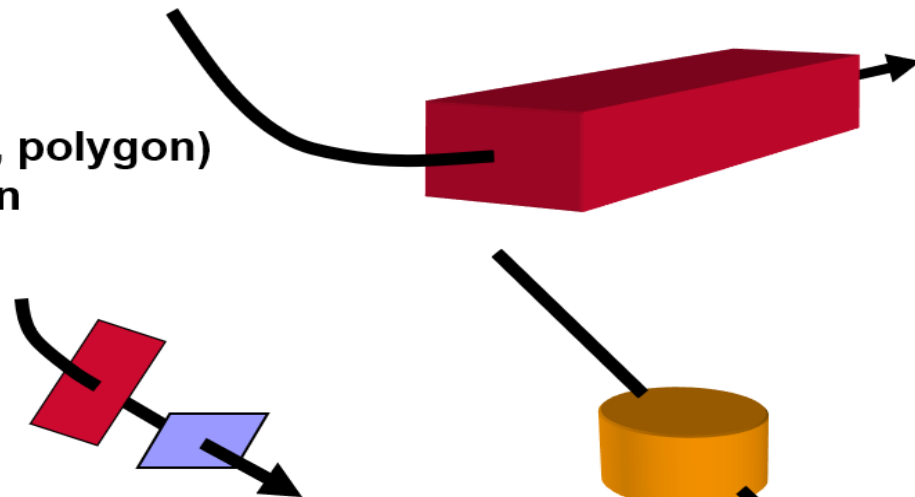
- Tools – Articulated MWD and Motors / Flex motors for short radius

What is a Target ?

- A defined area - or volume - to be penetrated by the well bore at a fixed location (given depth & coordinates)

What matters :

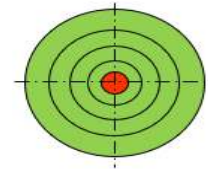
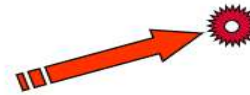
- Size (tolerance)
- Shape (circle, rectangle, polygon)
- Inclination at penetration
- Direction at penetration



Target Shape (1)

Point

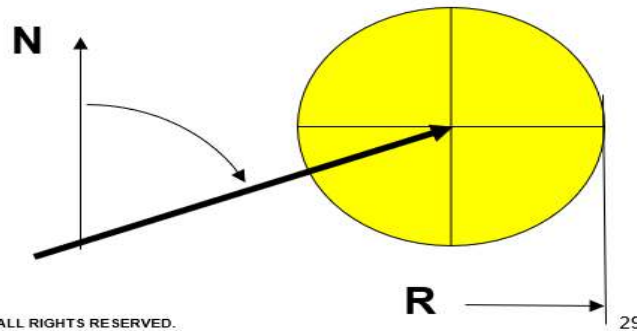
- coordinates
- inclination and direction into target



The Bull's Eye

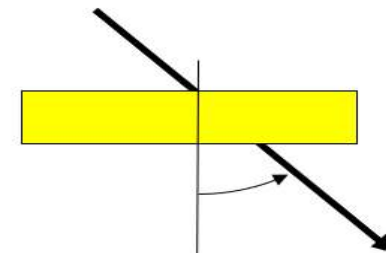
Circle

- coordinates of center
- radius or diameter
- inclination and direction into target



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29



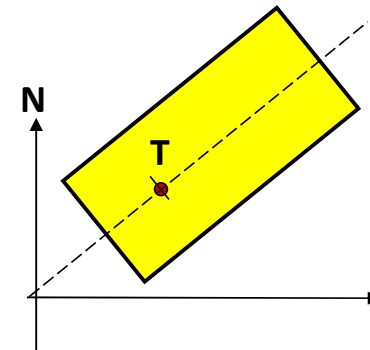
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Target Shape (2)

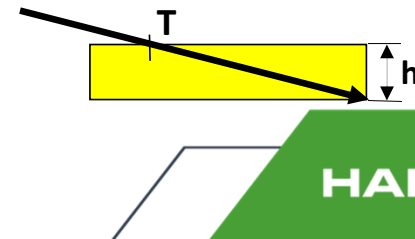
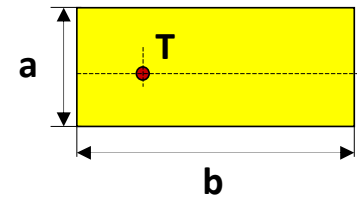
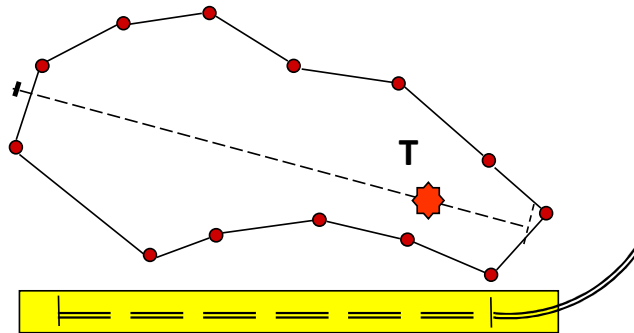
Rectangle

- coordinates of preferred aim point
- coordinates of one point
- orientation



Polygon

- coordinates of preferred aim point
- coordinates of corners



Geological Target Vs Driller Target



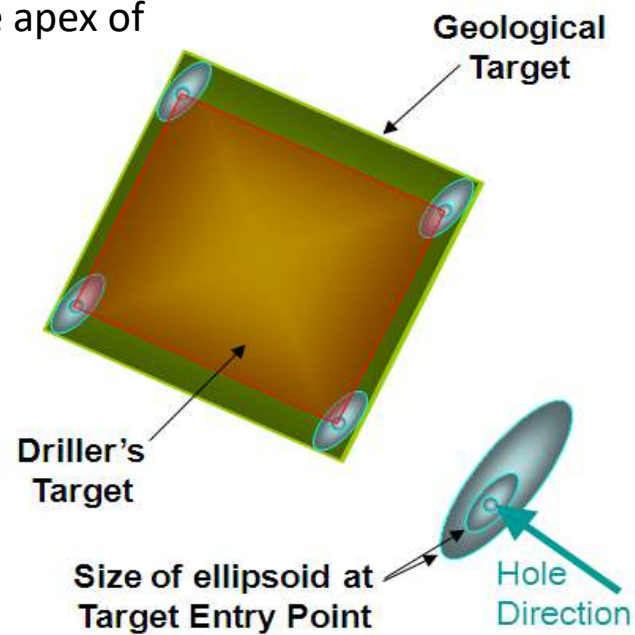
- Typically the geological target is described in term of a volume or area of rock the borehole must penetrate
 - In the case of a volume, there may also be a specific (minimum) length the wellbore must stay in the rock
- Due to survey uncertainty we can generally never be 100% certain that the geological target has been met. We can increase our confidence by:
 - Increasing the accuracy of the survey program
 - Require the directional driller to penetrate the target at the center
- Both of these options can have a significant cost so a compromise is the target analysis to determine the drillers target

Improving Driller's Target



Drilling target is defined by placing uncertainty ellipse at the apex of the Geological target and joining the penetration points

Smaller ellipses will give larger drillers targets



Bad planning means bad results



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Thank You!



Q&A

Sperry Drilling Svc.